Credit Name: CSE3010 - Object-Oriented Programming 2

Assignment Name: UEmployee, Faculty, Staff

How has your program changed from planning to coding to now? Please explain?

Understanding The Problem

The problem asked for a superclass that includes the variable fuelEconomyCty, fuelEconomyHwy, seatingCapacity and cargoVolume. The 3 subclasses should also have variables specific to its vehicle type. I was confused about how to effectively use abstract methods and why we would use them. I asked Mr. Abdulla and looked online to find some answers, resolving any confusion I had.

Planning The Solution/Implementation

I started with the superclass, including all the required variables and methods before thinking of things that all vehicles have to use the abstract method. I then made a Car subclass with variables specific to its vehicle type and returned the abstract method, repeating it for the Truck and Minivan subclass. In the client code, I created a method that returns an object. I prompted the user to input information relating to superclass, before asking the user for the type of vehicle. Based on the response, I prompted the user to input information specific to that vehicle type and returned an object using that information. I also used a while loop to continue prompting the user for a vehicle type until a valid input is entered. In the main method, I used a do-while loop to continue the program until "q" was entered. I also created 3 separate arraylist to organize the objects based on vehicle type. I first displayed the possible actions to the user and prompted the user to enter an action, using a while loop to repeat the prompt if an invalid input was entered. If the user wanted to add a vehicle to the catalogue, I called the object method I described above. I then used an accessor method from the superclass to get the vehicle type of the object to add it to its respective arraylist. If the user wanted to view the vehicles, I asked the user which type of vehicle they want to see before displaying all the objects in the repective arraylist using a for-loop. I also added a prompt that would display if the arraylist was empty. I had to regularly test the code everytime I finished a section to ensure that the loops worked as expected. If errors came up, I would use *System.out.print* to figure out where exactly it went wrong. Trial and error played a major part in fixing any errors.

Overcoming Challenges

The hardest part was figuring out what to use the abstract method for. I struggled with comprehending what exactly should be returned for this program. I talked to Mr. Abdalla and looked online before realizing that the answer was simplier than I thought. By using abstract methods, I can ensure that a vehicle's vehicleType(), vehicleModel(), and vehicleColour() is implemented in the subclasses.

Learning

While I am still a little unsure when and in what situations I should be using abstract methods in general, having the ability to do so will help me in the future.